OS & SP

Name : M Ramya Sree **Roll No :** 2520030094

SKILL – 22

1.Skill22.c

```
#!/bin/bash

echo "==================================="
echo " PROGRAM 22 AUTOMATED TEST"
echo "==================================="

echo ""
echo "Test 1: Checking executable"

if [ -f "./program22" ]; then
    echo "PASS: Executable exists"
else
    echo "FAIL: Executable not found"
```

```
    exit 1
fi

echo ""
echo "Test 2: Running program"

./program22 > output.txt

if [ $? -eq 0 ]; then
    echo "PASS: Program executed successfully"
else
    echo "FAIL: Program execution failed"
    exit 1
fi

echo ""
echo "Test 3: Checking output"
```

```
if grep -q "All tests completed." output.txt; then
    echo "PASS: Expected output found"
else
    echo "FAIL: Expected output missing"
    exit 1
fi

echo ""
echo "Test 4: Checking memory"

valgrind --leak-check=full ./program22 > valgrind.txt 2>&1

if grep -q "All heap blocks were freed" valgrind.txt; then
    echo "PASS: No memory leak detected"
else
    echo "FAIL: Memory leak may exist"
fi
```

echo ""

echo "======================================"

echo " TESTING COMPLETED"

echo "======================================"

OUTPUT :

```
ramyasree@DESKTOP-DP2O9U6:~/Skill$ nano skill22.c
ramyasree@DESKTOP-DP2O9U6:~/Skill$ gcc skill22.c -o skill22
ramyasree@DESKTOP-DP2O9U6:~/Skill$ chmod +x skill22.sh
./skill22.sh
==================================
 PROGRAM 22 AUTOMATED TEST
==================================

Test 1: Checking executable
FAIL: Executable not found
ramyasree@DESKTOP-DP2O9U6:~/Skill$ cat > program22.c <<'EOF'
#include <stdio.h>

int main()
{
    printf("All tests completed.\n");
    return 0;
}
EOF
ramyasree@DESKTOP-DP2O9U6:~/Skill$ gcc program22.c -o program22
bash skill22.sh
==================================
 PROGRAM 22 AUTOMATED TEST
==================================

Test 1: Checking executable
PASS: Executable exists

Test 2: Running program
PASS: Program executed successfully

Test 3: Checking output
PASS: Expected output found

Test 4: Checking memory
FAIL: Memory leak may exist

==================================
 TESTING COMPLETED
==================================
ramyasree@DESKTOP-DP2O9U6:~/Skill$
```